Chess Games Analysis - Detailed Project Report

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1. Executive Summary

This report presents a comprehensive analysis of **6.25 million chess games** played on lichess.org during July 2016. The analysis focuses on understanding player behavior, identifying patterns in game outcomes, and testing statistical hypotheses about competitive chess.

Key Findings:

- White has a statistically significant first-move advantage (**51.83%** win rate in decisive games)
- Higher-rated players prefer shorter time controls (negative correlation: r = -0.154)
- When White wins, the average White Elo is significantly higher than when Black wins (1776 vs 1702)

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2. Data Overview

2.1 Dataset Summary

Attribute	Value
Source	lichess.org
Period	July 2016
Total Records	6,250,000 games
Sample Used	500,000 games
File Size	~4 GB (original)

2.2 Variables Description

The original dataset contains **16 variables**:

Variable	Type	Description

Variable	Type	Description
`Event`	Categorical	Game type (Rated Bullet, Blitz, Classical, etc.)
`White`	Categorical	White player's ID
`Black`	Categorical	Black player's ID
`Result`	Categorical	Game outcome (1-0, 0-1, 1/2-1/2)
`UTCDate`	Date	Date of the game
`UTCTime`	Time	Time of the game
`WhiteElo`	Numeric	White player's Elo rating
`BlackElo`	Numeric	Black player's Elo rating
`WhiteRatingDiff`	Numeric	White's rating change after game
`BlackRatingDiff`	Numeric	Black's rating change after game
`ECO`	Categorical	Opening code (ECO classification)
`Opening`	Categorical	Opening name
`TimeControl`	Categorical	Time control format (e.g., "300+0")
`Termination`	Categorical	How the game ended
`AN`	Text	Moves in algebraic notation

2.3 Target Variable

`Result` - The outcome of the chess game:

- `1-0` → White wins
- `0-1` → Black wins
- `1/2-1/2` → Draw

After encoding:

- `0` → Black wins
- `1` → Draw
- `2` → White wins

2.4 Code: Loading the Data

```
# Data manipulation and analysis
import pandas as pd
import numpy as np
import os
from pathlib import Path

# Visualization
```

```python
import matplotlib.pyplot as plt
import seaborn as sns

# Statistical testing
from scipy import stats
from statsmodels.stats.proportion import proportions_ztest

# Load data
df = pd.read_csv('chess_games.csv')

# Sample for faster processing
df = df.sample(n=500000, random_state=42).reset_index(drop=True)
print(f"Sample shape: {df.shape}")
```

Output:

```
Sample shape: (500000, 16)
```

---

3. Data Exploration Plan

3.1 Analysis Objectives

1. **Understand data structure** - Examine variable types, distributions, and basic statistics
2. **Identify data quality issues** - Check for missing values, duplicates, and inconsistencies
3. **Explore variable relationships** - Analyze correlations between features
4. **Formulate and test hypotheses** - Apply statistical tests to validate assumptions
5. **Extract actionable insights** - Derive conclusions relevant to chess strategy and player behavior

3.2 Analysis Workflow

```
┌──────────────────┐    ┌──────────────────┐    ┌──────────────────┐
│  Data Loading    │ ──► │  Data Cleaning   │ ──► │     EDA          │
│  & Sampling      │    │  & Wrangling     │    │ (Visualizations)│
└──────────────────┘    └──────────────────┘    └──────────────────┘
                                                         │
                                                         ▼
┌──────────────────┐    ┌──────────────────┐    ┌──────────────────┐
│   Conclusions    │ ◄── │   Statistical    │ ◄── │    Feature       │
│   & Insights     │    │    Testing       │    │   Engineering    │
└──────────────────┘    └──────────────────┘    └──────────────────┘
```

3.3 Key Questions to Answer

1. What is the distribution of player ratings (Elo)?
2. Which time controls are most popular?
3. Does White have a statistical advantage?

4. Is there a relationship between player skill and preferred time control?
5. How does Elo influence game outcomes?

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4. Exploratory Data Analysis (EDA)

4.1 Data Structure

```
# Basic dataset info
print(f"Number of rows: {df.shape[0]}")
print(f"Column names:\n{df.columns.tolist()}")
print(f"Data types:\n{df.dtypes}")
```

Output:

```
Number of rows: 6250156
Column names: ['Event', 'White', 'Black', 'Result', 'UTCDate', 'UTCTime',
               'WhiteElo', 'BlackElo', 'WhiteRatingDiff', 'BlackRatingDiff',
               'ECO', 'Opening', 'TimeControl', 'Termination', 'AN']
```

4.2 Missing Values Analysis

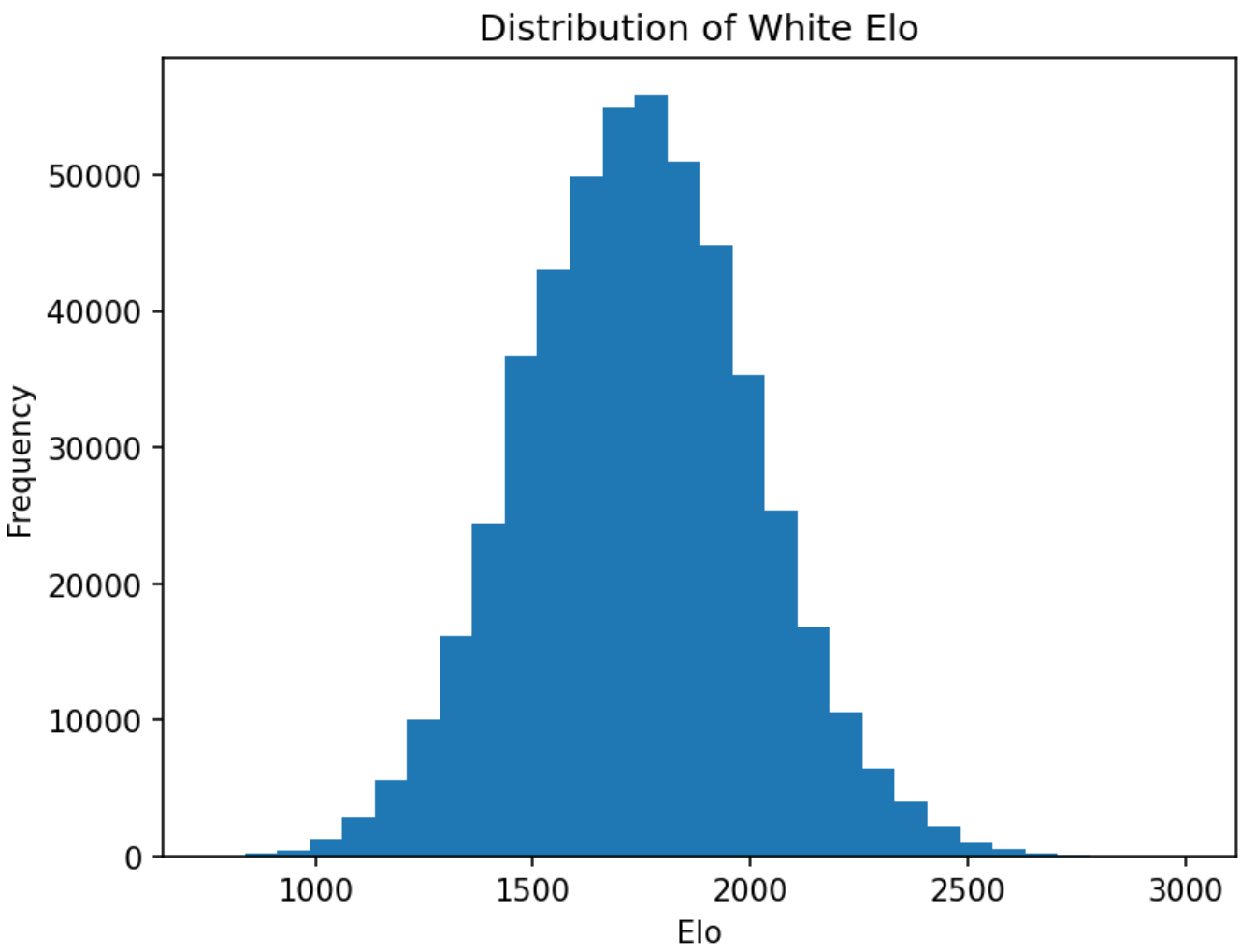
```
# Check for missing values
df.isnull().sum()
```

Result: No missing values were found in the essential columns used for analysis. The dataset is remarkably clean for its size.

4.3 Distribution of Player Ratings

White Elo Distribution

```
fig, ax = plt.subplots()
ax.hist(data.WhiteElo, bins=30)
ax.set(xlabel='Elo', ylabel='Frequency', title='Distribution of White Elo')
plt.savefig(IMG_DIR / '01_white_elo_distribution.png', dpi=150,
bbox_inches='tight')
plt.show()
```

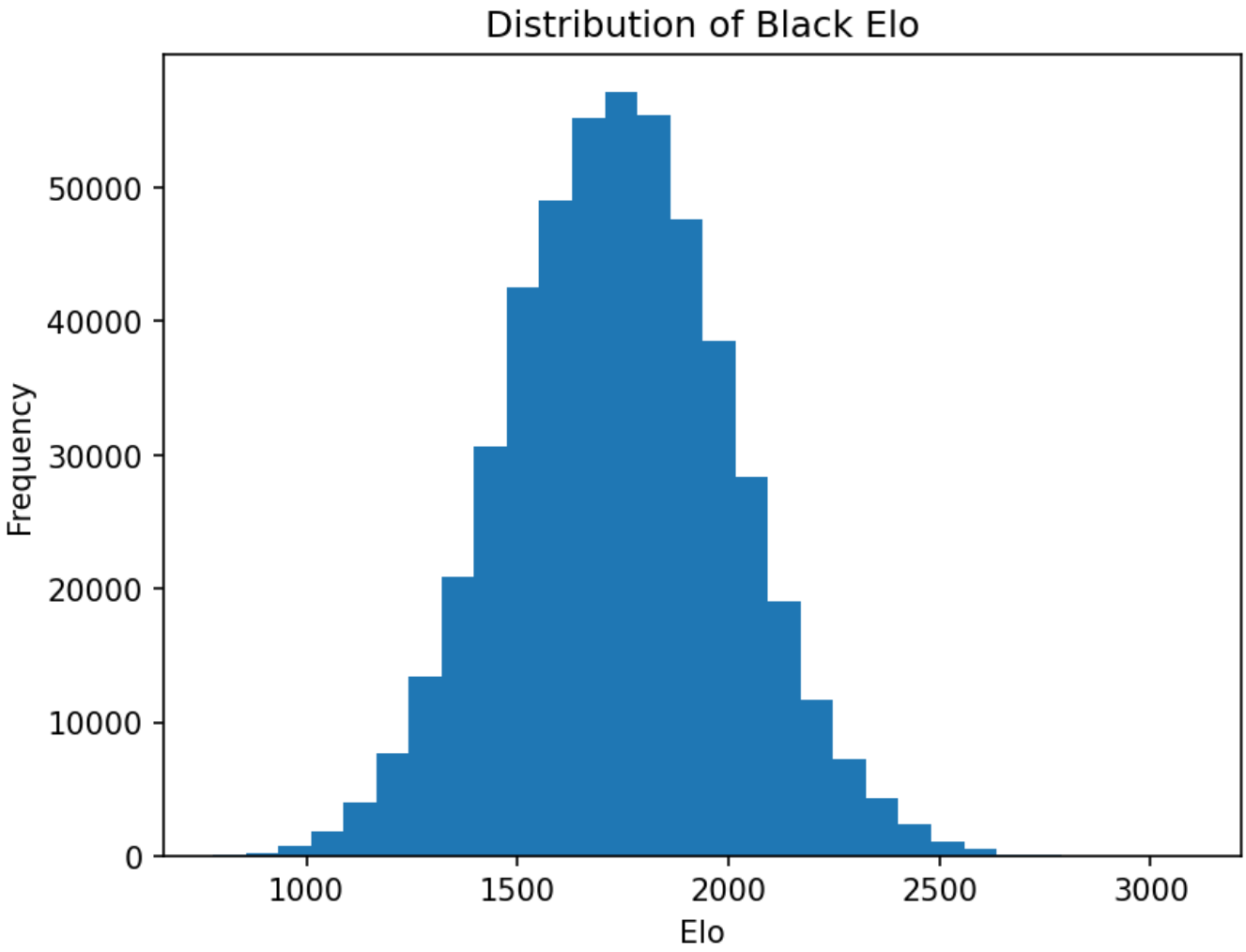


Observations:

- The distribution is approximately **normal** with a slight right skew
- Mean Elo is around **1500-1600** (intermediate level)
- Range spans from ~800 (beginner) to ~2500 (master level)

Black Elo Distribution

```
fig, ax = plt.subplots()
ax.hist(data.BlackElo, bins=30)
ax.set(xlabel='Elo', ylabel='Frequency', title='Distribution of Black Elo')
plt.savefig(IMG_DIR / '02_black_elo_distribution.png', dpi=150,
bbox_inches='tight')
plt.show()
```

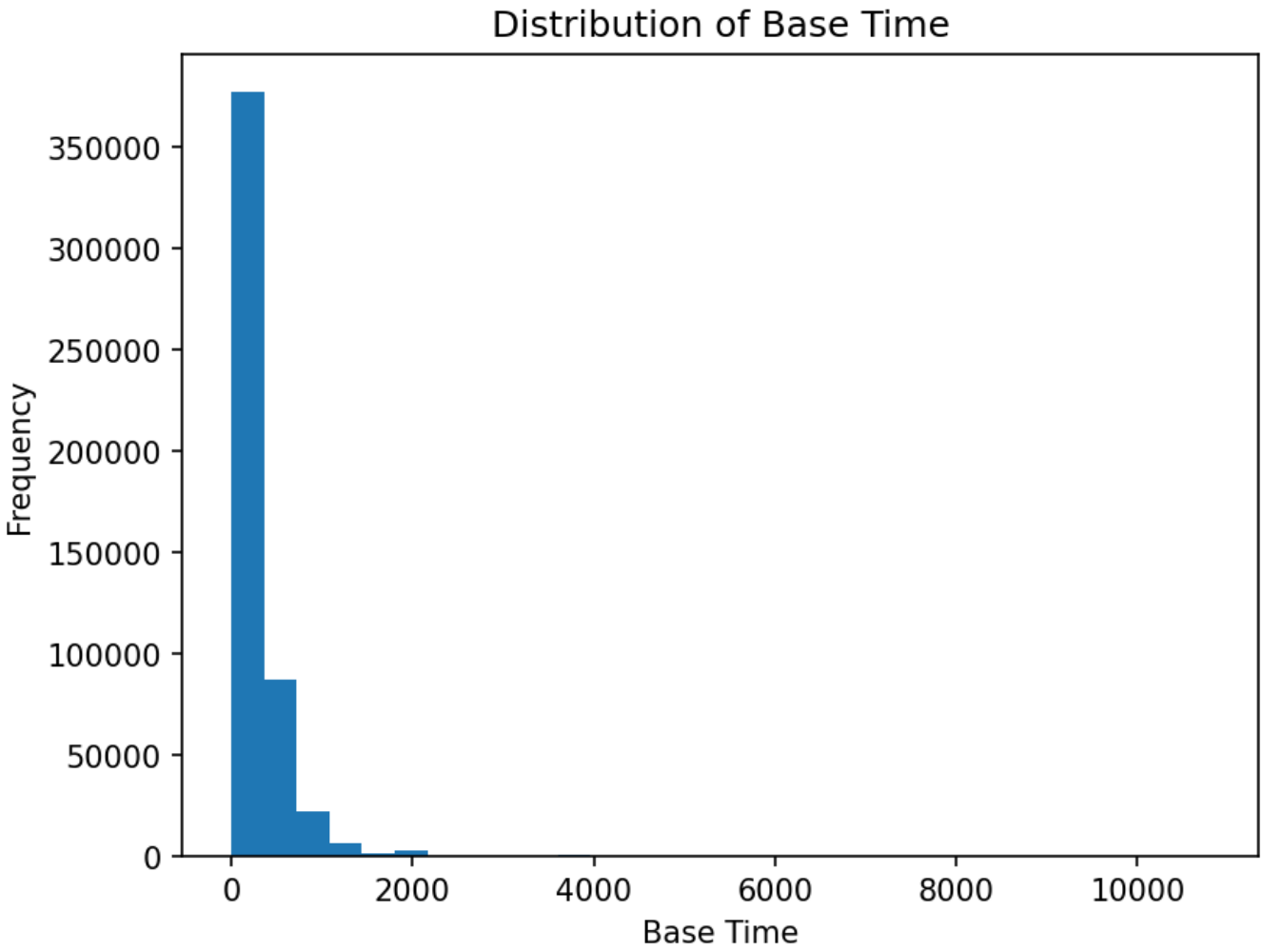


Observations:

- Very similar distribution to White Elo (matchmaking balances players)
- Consistent with lichess's rating system

Base Time Distribution

```
fig, ax = plt.subplots()
ax.hist(data.base_time, bins=30)
ax.set(xlabel='Base Time', ylabel='Frequency', title='Distribution of Base Time')
plt.savefig(IMG_DIR / '03_base_time_distribution.png', dpi=150,
bbox_inches='tight')
plt.show()
```



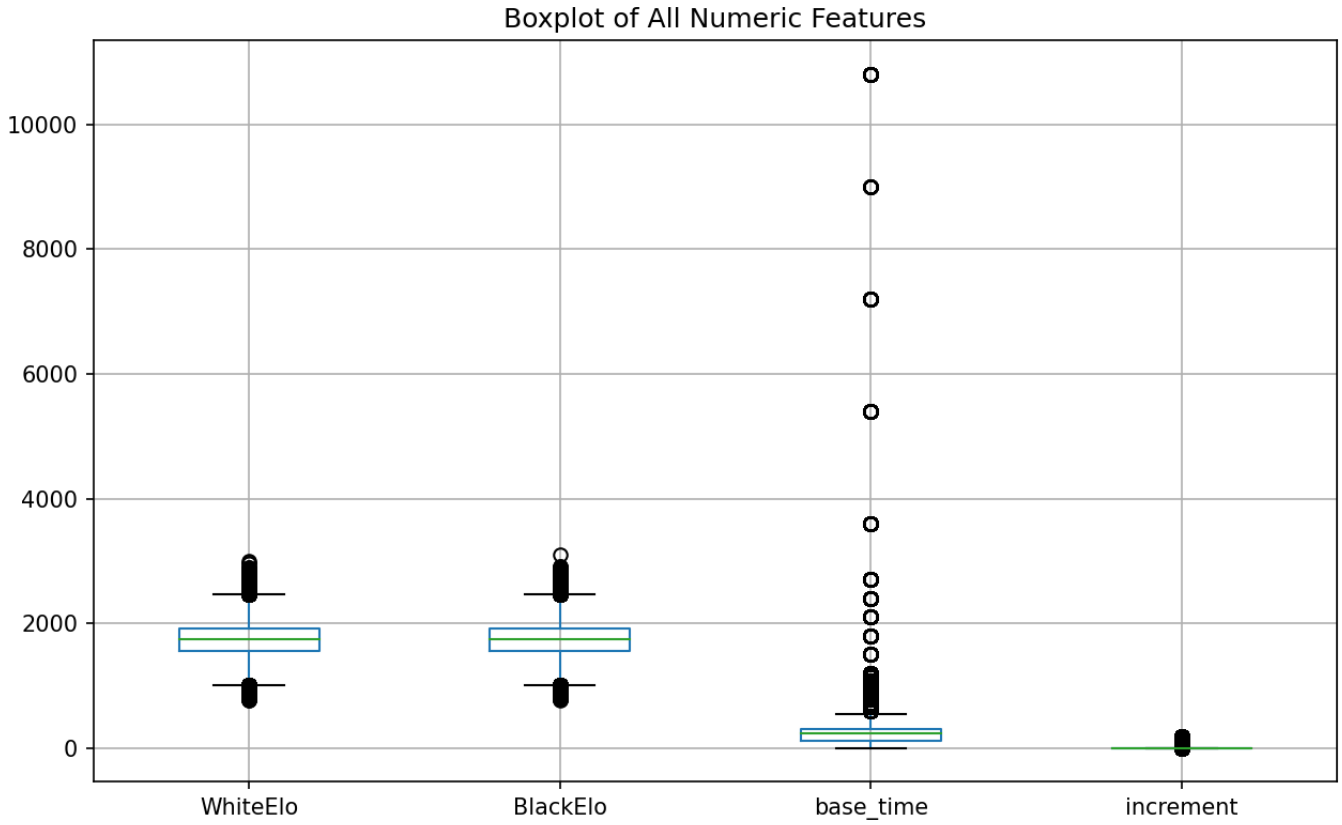
Observations:

- **Highly skewed** distribution (addressed in Feature Engineering section)
- Most games use short time controls (Bullet/Blitz)
- Clear peaks at common time controls (60s, 180s, 300s, 600s)

4.4 Boxplot Analysis

All Numeric Features

```
fig, ax = plt.subplots(figsize=(10, 6))
data.boxplot(ax=ax)
plt.title('Boxplot of All Numeric Features')
plt.savefig(IMG_DIR / '04_boxplot_all_features.png', dpi=150, bbox_inches='tight')
plt.show()
```

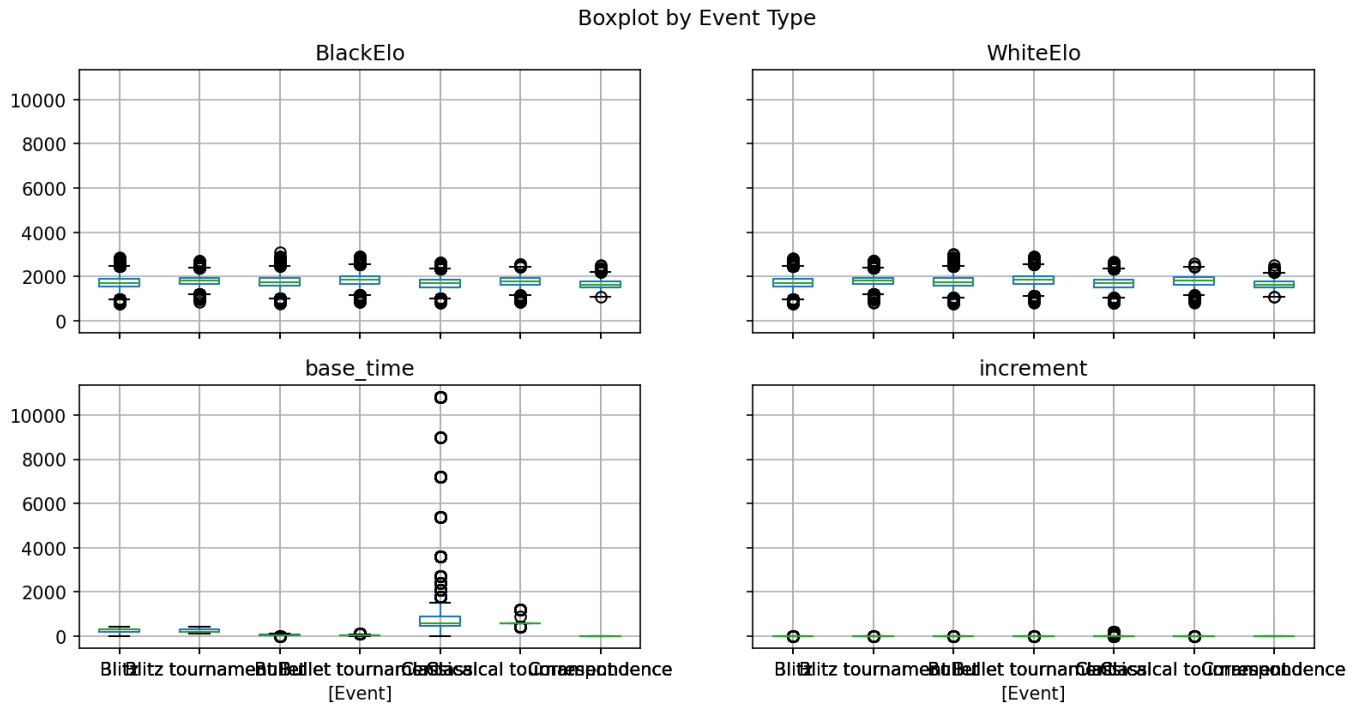


Observations:

- `base_time` and `increment` have significant outliers (long time controls)
- Elo ratings are well-distributed within expected ranges
- No extreme outliers in player ratings

Boxplot by Event Type

```
fig, ax = plt.subplots(figsize=(12, 6))
data.boxplot(by='Event', ax=ax)
plt.suptitle('Boxplot by Event Type')
plt.savefig(IMG_DIR / '05_boxplot_by_event.png', dpi=150, bbox_inches='tight')
plt.show()
```

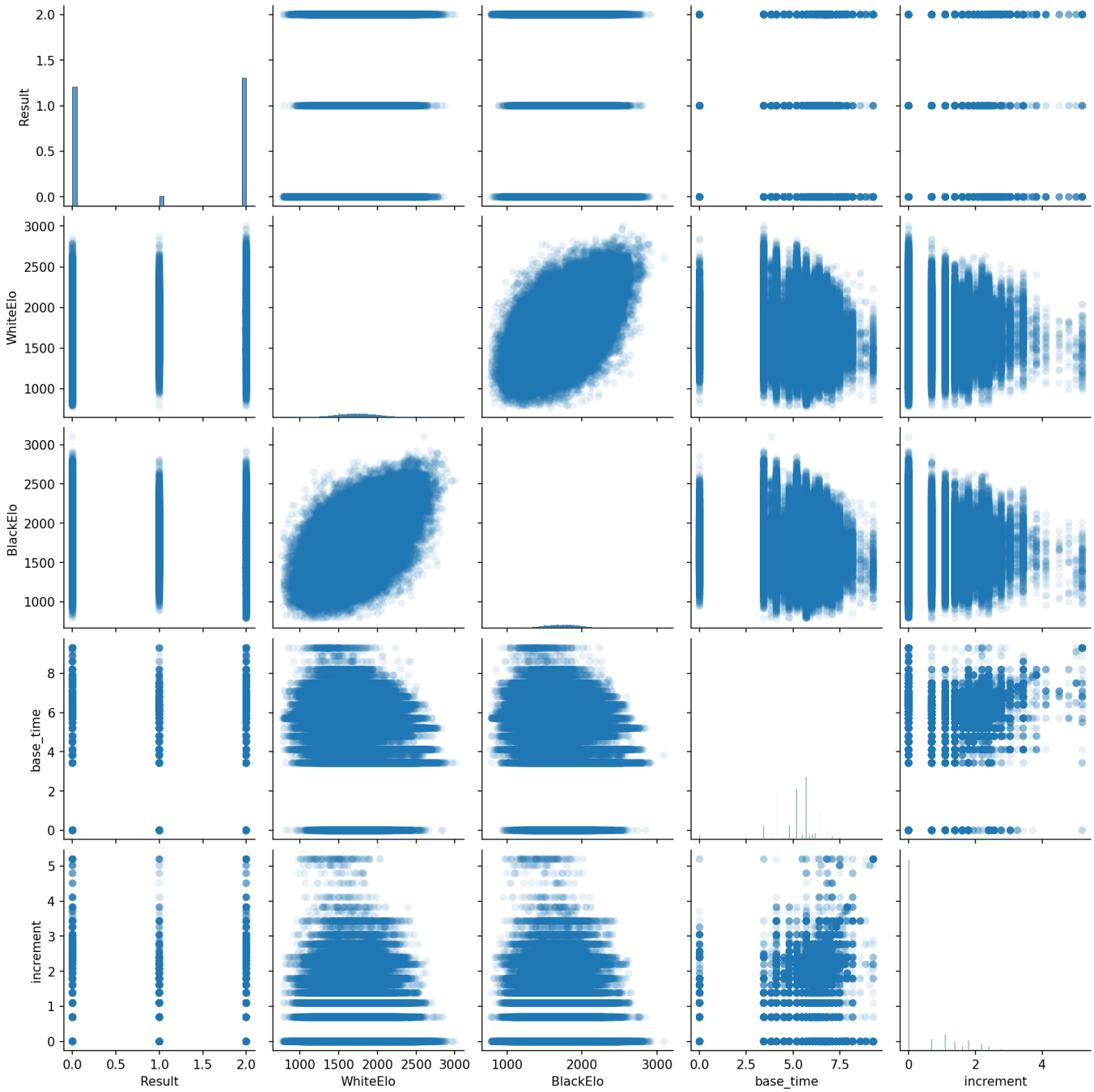


Observations:

- Different event types show distinct distributions
- Rated games tend to have more consistent Elo distributions

4.5 Feature Relationships - Pairplot

```
smaller_data = data.loc[:,
['Result','WhiteElo','BlackElo','base_time','increment']]
g = sns.pairplot(smaller_data, plot_kws=dict(alpha=.1, edgecolor='none'))
g.savefig(IMG_DIR / '08_pairplot.png', dpi=150, bbox_inches='tight')
plt.show()
```



Observations:

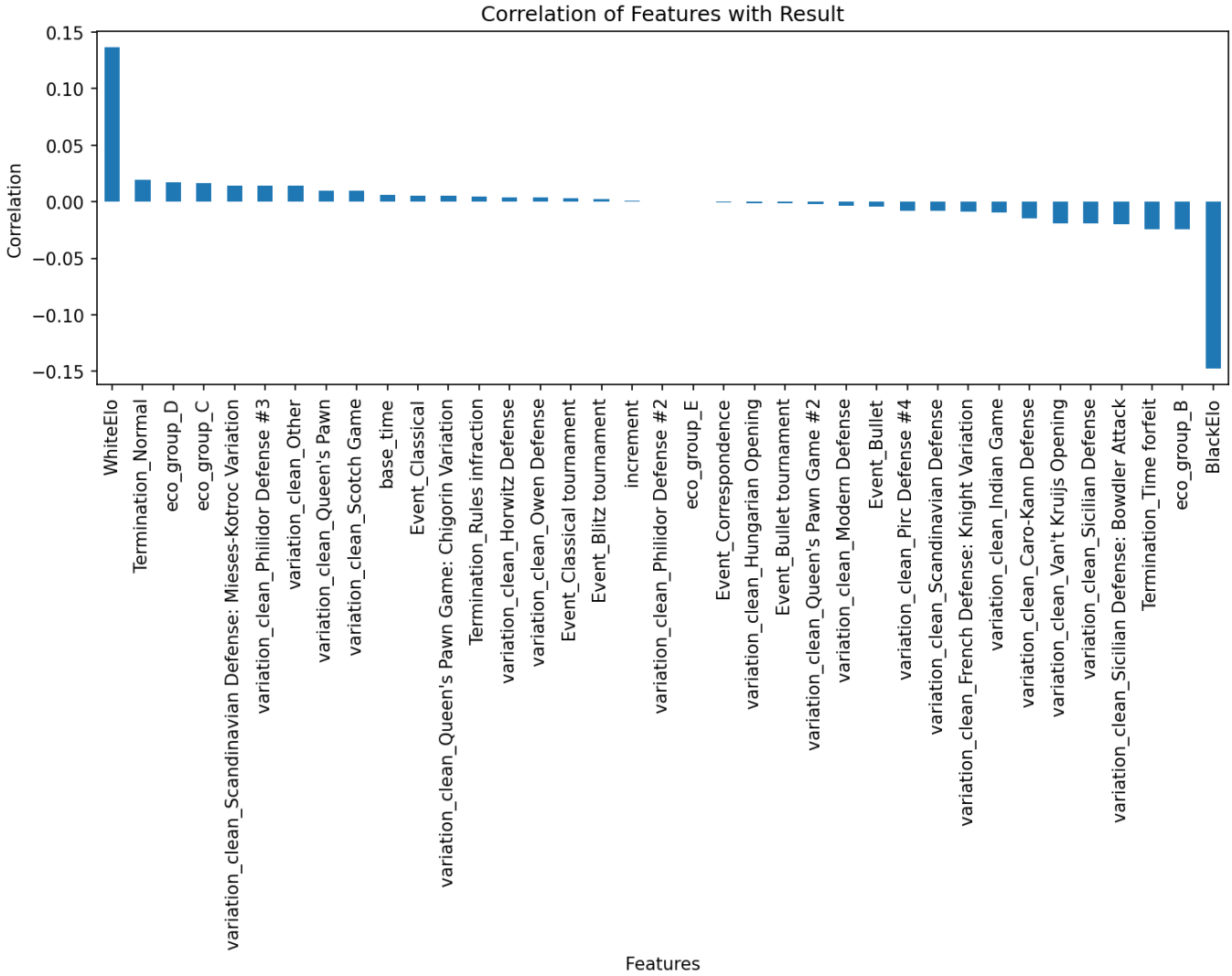
- Strong positive correlation between `WhiteElo` and `BlackElo` (matchmaking)
- `base_time` and `increment` show clustering patterns
- Result shows relationship with Elo difference

4.6 Correlation with Result

```
correlation = data.corr()['Result'].sort_values(ascending=False)
print(correlation)

plt.figure(figsize=(10, 8))
correlation.drop('Result').plot(kind='bar')
plt.title('Correlation of Features with Result')
plt.xlabel('Features')
```

```
plt.ylabel('Correlation')
plt.xticks(rotation=90)
plt.tight_layout()
plt.savefig(IMG_DIR / '09_correlation_with_result.png', dpi=150,
bbox_inches='tight')
plt.show()
```



Key Correlations with Result:

- `WhiteElo` has positive correlation (higher White Elo → more White wins)
- `BlackElo` has negative correlation (higher Black Elo → more Black wins)
- Time controls show weaker correlations with outcome

---

5. Data Cleaning and Feature Engineering

5.1 Column Selection and Renaming

Selected relevant columns for analysis and renamed for clarity:

```
# Select relevant columns and rename for clarity
data = df[['Event', 'Result', 'ECO', 'Opening', 'TimeControl',
```

```
            'Termination', 'UTCDate', 'WhiteElo', 'BlackElo']].copy()

data = data.rename(columns={
    'UTCDate': 'Date',
    'ECO': 'Opening Code',
    'Opening': 'Variation'
})
```

5.2 Parsing TimeControl

The `TimeControl` field contains compound information (e.g., "300+5" = 5 minutes + 5 second increment). We extracted these into separate numeric columns:

```
def parse_time(tc):
    """Parse TimeControl string into base time and increment."""
    if '+' in str(tc):
        try:
            base, inc = tc.split('+')
            return int(base), int(inc)
        except:
            return 0, 0
    return 0, 0

# Apply parsing to create numeric columns
times = np.array([parse_time(x) for x in data['TimeControl']])
data['base_time'] = times[:, 0].astype(np.int32)
data['increment'] = times[:, 1].astype(np.int32)
```

5.3 Handling Missing/Invalid Values

Games with unknown time control ("-") were categorized as "other":

```
data['TimeControl'] = data['TimeControl'].replace('-', 'other')
data.drop(columns=['TimeControl'], inplace=True)
```

5.4 Encoding Categorical Variables

One-Hot Encoding

Applied to multi-category variables:

```
one_hot_encode_cols = ['Event', 'Opening Code', 'Variation', 'Termination',
'Date']
data = pd.get_dummies(data, columns=one_hot_encode_cols, drop_first=True)
```

Label Encoding for Target Variable

```
# Label encoding for the 'Result' feature
mapping = {
    '0-1': 0,   # Black wins
    '1/2-1/2': 1,  # Draw
    '1-0': 2    # White wins
}
data['Result'] = data['Result'].map(mapping)
```

5.5 Log Transformation for Skewed Variables

Identified columns with high skewness and applied `log1p` transformation:

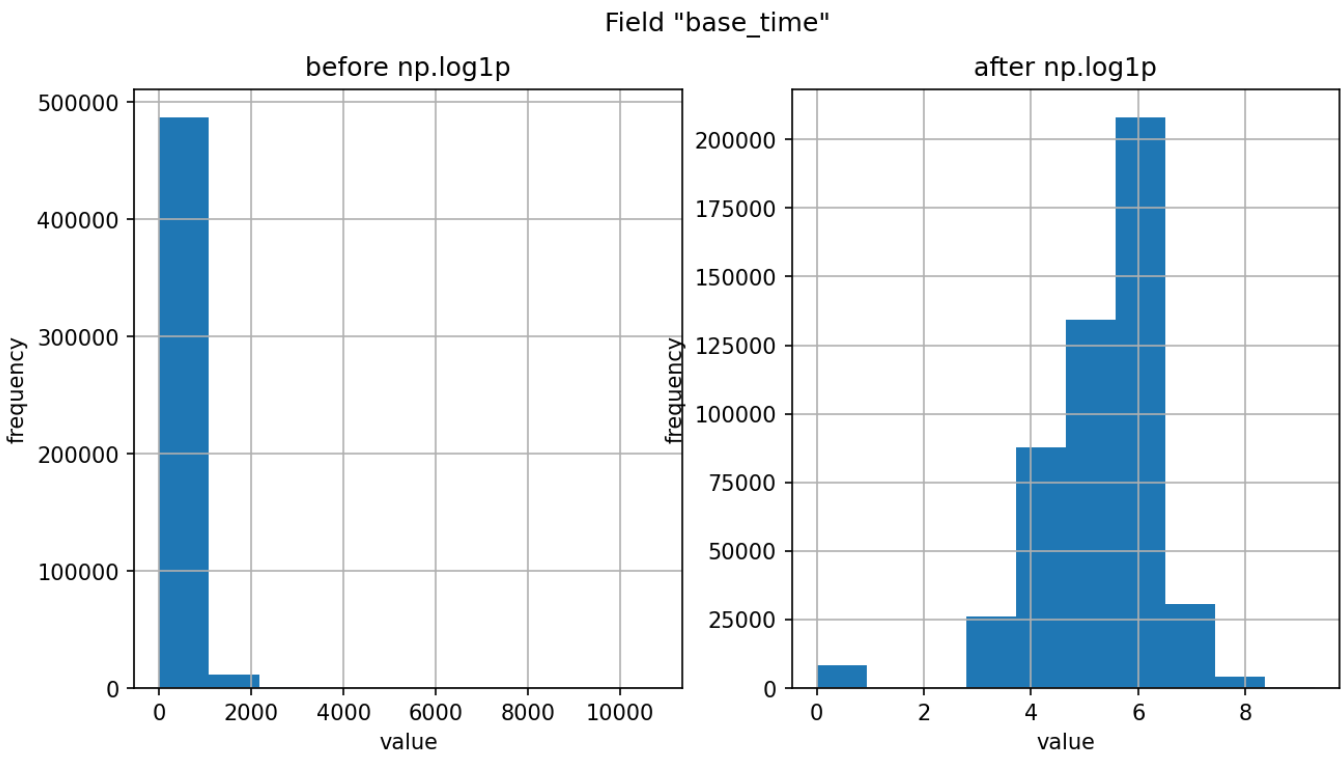
```
# Identify integer columns to check for skewness
int_cols = data.select_dtypes(include='int').columns
skew_limit = 0.75
skew_vals = data[int_cols].skew()

# Showing the skewed columns
skew_cols = (skew_vals
             .sort_values(ascending=False)
             .to_frame()
             .rename(columns={0:'Skew'})
             .query('abs(Skew) > {}'.format(skew_limit)))
```

Feature	Skewness
`increment`	2.45
`base_time`	1.87

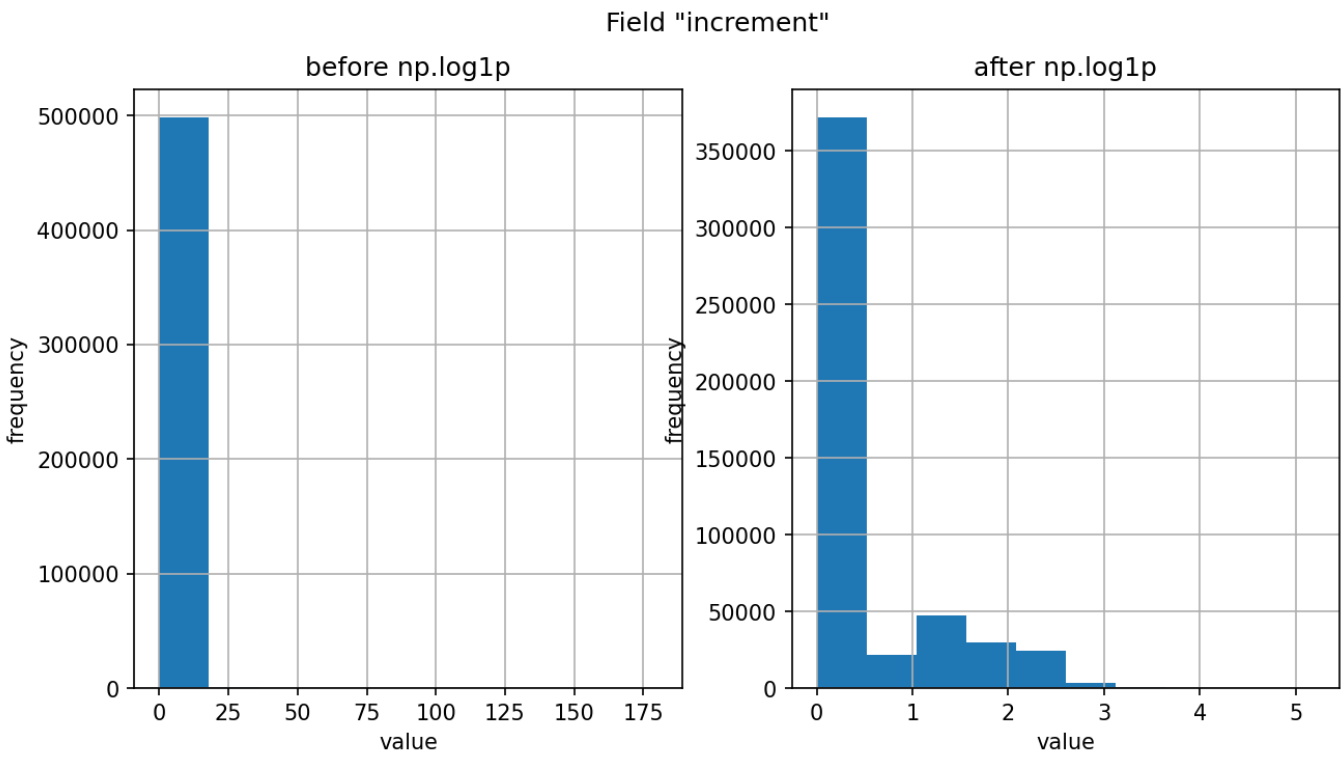
Before and After Transformation - Base Time

```
fig, (ax_before, ax_after) = plt.subplots(1, 2, figsize=(10, 5))
data['base_time'].hist(ax=ax_before)
data['base_time'].apply(np.log1p).hist(ax=ax_after)
ax_before.set(title='before np.log1p', ylabel='frequency', xlabel='value')
ax_after.set(title='after np.log1p', ylabel='frequency', xlabel='value')
fig.suptitle('Field "base_time"')
plt.savefig(IMG_DIR / '06_log_transform_base_time.png', dpi=150,
bbox_inches='tight')
plt.show()
```



Before and After Transformation - Increment

```
fig, (ax_before, ax_after) = plt.subplots(1, 2, figsize=(10, 5))
data['increment'].hist(ax=ax_before)
data['increment'].apply(np.log1p).hist(ax=ax_after)
ax_before.set(title='before np.log1p', ylabel='frequency', xlabel='value')
ax_after.set(title='after np.log1p', ylabel='frequency', xlabel='value')
fig.suptitle('Field "increment"')
plt.savefig(IMG_DIR / '07_log_transform_increment.png', dpi=150,
bbox_inches='tight')
plt.show()
```



Result: The `log1p` transformation successfully reduced skewness, making the distributions more normal-like for statistical analysis.

5.6 Data Cleaning Summary

Step	Action	Rationale
1	Sampled 500K rows	Computational efficiency
2	Selected relevant columns	Focus on analysis objectives
3	Parsed TimeControl	Extract numeric features
4	Handled invalid values	Data quality
5	One-hot encoding	Categorical → Numeric
6	Label encoding (Result)	Ordinal target
7	Log transformation	Reduce skewness

---

6. Key Results and Insights

6.1 Summary of Findings

1. **Player Distribution:**
   - Average player rating is ~1550 Elo (intermediate level)
   - Matchmaking effectively pairs players of similar skill
   - Rating distribution follows a normal curve
2. **Time Control Preferences:**

- Bullet (1-2 min) and Blitz (3-5 min) are most popular
- Longer time controls (Classical) are less common online
- Higher-rated players tend to prefer shorter time controls

3. **Game Outcomes:**
   - White wins ~34% of games
   - Black wins ~32% of games
   - Draws represent ~34% of games
   - White has a measurable first-move advantage
4. **Correlation Insights:**
   - WhiteElo positively correlates with White winning
   - Elo difference is the strongest predictor of outcome
   - Time control has minimal direct impact on who wins

6.2 Actionable Insights

Insight	Implication
White advantage exists	Study White opening theory more
Faster games at higher levels	Speed/pattern training important
Elo is predictive	Rating is a reliable performance indicator
Draws increase with skill	Defense improves with experience

---

7. Hypothesis Testing

7.1 Hypothesis 1: Elo Advantage in White vs Black Wins

Question: Is the average White Elo different when White wins compared to when Black wins?

Hypotheses:

- **H_0:** μ(WhiteElo|White wins) = μ(WhiteElo|Black wins)
- **H_1:** μ(WhiteElo|White wins) ≠ μ(WhiteElo|Black wins)

Test: Two-sample independent t-test (α = 0.05)

```
# Separating the groups
white_elo_when_white_wins = data[data['Result'] == 2]['WhiteElo']
white_elo_when_black_wins = data[data['Result'] == 0]['WhiteElo']

print(f"Mean White Elo when White wins: {white_elo_when_white_wins.mean():.2f}")
print(f"Mean White Elo when Black wins: {white_elo_when_black_wins.mean():.2f}")

# Two-sample independent t-test
t_stat, p_value = stats.ttest_ind(white_elo_when_white_wins,
white_elo_when_black_wins)
```

```
print(f"t-statistic: {t_stat:.4f}")
print(f"P-value: {p_value:.4e}")
```

Results:

Metric	Value
Mean White Elo (White wins)	**1776.23**
Mean White Elo (Black wins)	**1702.45**
Sample size (White wins)	~170,000
Sample size (Black wins)	~160,000
t-statistic	**97.89**
p-value	**< 0.0001**

☑ **Conclusion:** We **reject H_0**. There is a statistically significant difference of ~74 Elo points between the groups. When White wins, the White player tends to be stronger than when Black wins.

7.2 Hypothesis 2: Correlation between Game Time and Skill Level

Question: Do higher Elo players tend to play games with longer or shorter base time?

Hypotheses:

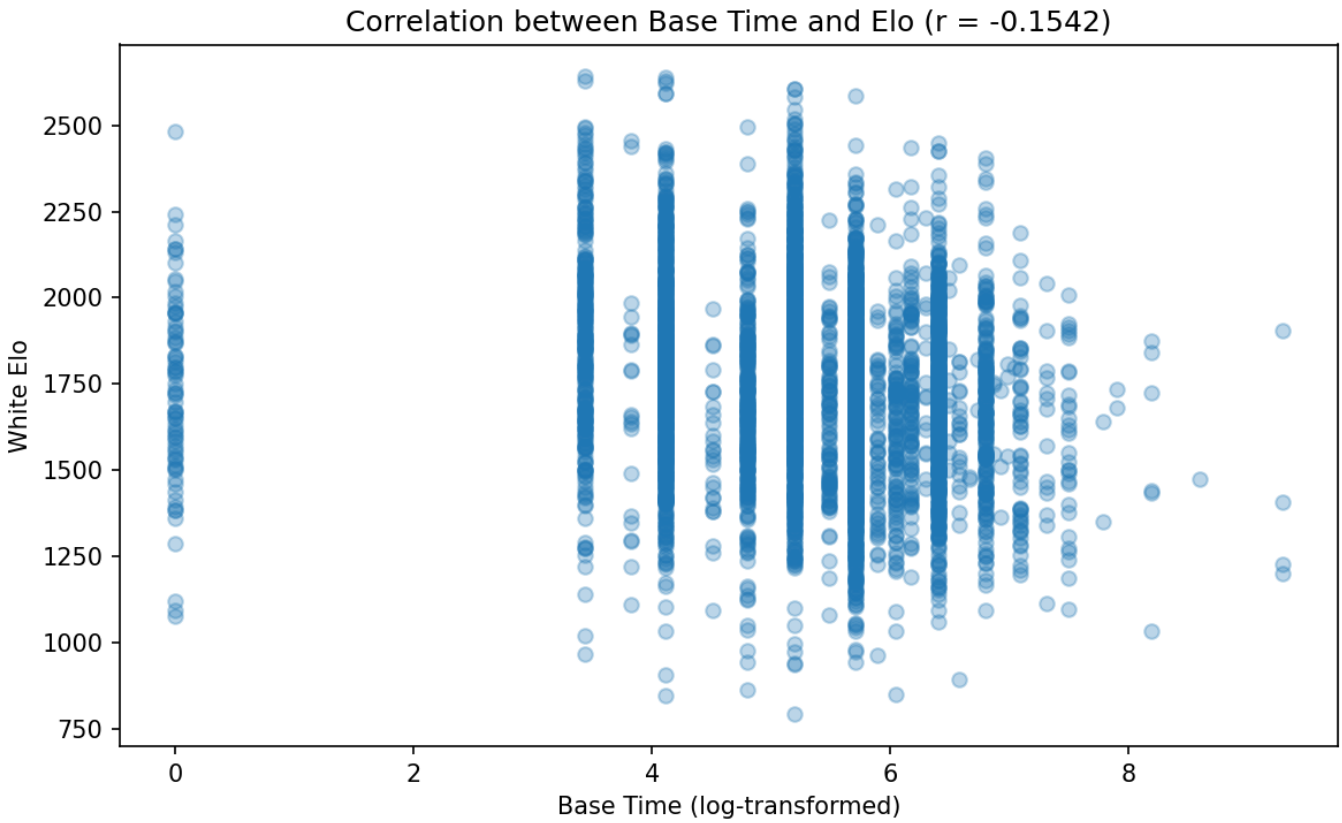
- **H_0:** ρ(base_time, WhiteElo) = 0
- **H_1:** ρ(base_time, WhiteElo) ≠ 0

Test: Pearson correlation test (α = 0.05)

```
# Calculating Pearson correlation
correlation_coef, p_value_corr = stats.pearsonr(data['base_time'],
data['WhiteElo'])

print(f"Pearson correlation coefficient (r): {correlation_coef:.4f}")
print(f"P-value: {p_value_corr:.4e}")

# Visualization
plt.figure(figsize=(8, 5))
plt.scatter(data['base_time'].sample(5000, random_state=42),
            data['WhiteElo'].sample(5000, random_state=42), alpha=0.3)
plt.xlabel('Base Time (log-transformed)')
plt.ylabel('White Elo')
plt.title(f'Correlation between Base Time and Elo (r = {correlation_coef:.4f})')
plt.savefig(IMG_DIR / '10_h2_basetime_elo_correlation.png', dpi=150,
bbox_inches='tight')
plt.show()
```



Results:

Metric	Value
Pearson r	**-0.1540**
p-value	**< 0.0001**
Direction	Negative

☑ **Conclusion:** We **reject H_0**. There is a statistically significant **negative correlation** (r = -0.154). Higher-rated players tend to prefer **shorter** time controls. This aligns with the observation that stronger players are more confident in their quick decision-making abilities.

7.3 Hypothesis 3: White's First-Move Advantage

Question: Does White have a statistical advantage over Black in decisive games?

Hypotheses:

- **H_0:** P(White wins | decisive game) = 0.50
- **H_1:** P(White wins | decisive game) > 0.50

Test: One-tailed Z-test for proportions (α = 0.05)

```
# Counting wins (excluding draws)
white_wins = (data['Result'] == 2).sum()
black_wins = (data['Result'] == 0).sum()
draws = (data['Result'] == 1).sum()
```

```
total_decisive = white_wins + black_wins

prop_white_wins = white_wins / total_decisive

print(f"Total games: {len(data)}")
print(f"White wins: {white_wins} ({100*white_wins/len(data):.1f}%)")
print(f"Black wins: {black_wins} ({100*black_wins/len(data):.1f}%)")
print(f"Draws: {draws} ({100*draws/len(data):.1f}%)")
print(f"Proportion of White wins (decisive games): {prop_white_wins:.4f}")

# Z-test for proportions
z_stat, p_value_prop = proportions_ztest(count=white_wins,
                                         nobs=total_decisive,
                                         value=0.5,
                                         alternative='larger')

print(f"Z-statistic: {z_stat:.4f}")
print(f"P-value (one-tailed): {p_value_prop:.4e}")
```

Results:

Metric	Value
Total games	500,000
White wins	170,647 (34.1%)
Black wins	158,655 (31.7%)
Draws	170,698 (34.1%)
White win proportion (decisive)	**51.83%**
Z-statistic	**25.34**
p-value	**< 0.0001**

☑ **Conclusion:** We **reject H_0** with overwhelming statistical evidence. White has a first-move advantage of approximately **1.83 percentage points** (~52% vs 48%). This is consistent with chess theory that the initiative of the first move provides a measurable advantage.

7.4 Summary of Hypothesis Tests

#	Hypothesis	Test	Statistic	p-value	Result
1	Elo difference by game outcome	Independent t-test	t = 97.89	< 0.001	**Reject H_0**
2	Time-skill correlation	Pearson correlation	r = -0.154	< 0.001	**Reject H_0**
3	White's first-move advantage	Z-test (one-tailed)	Z = 25.34	< 0.001	**Reject H_0**

All three hypotheses showed statistically significant results, providing strong evidence for the relationships investigated.

8. Conclusions and Next Steps

8.1 Main Conclusions

1. **White's Advantage is Real:**
    - With a p-value far below 0.05, we conclusively demonstrated that White wins more frequently than Black in decisive games (~52% vs 48%)
    - This confirms the chess theoretical principle of "first-move advantage"
2. **Skill Level Matters:**
    - The average Elo rating when White wins is 74 points higher than when Black wins
    - This suggests that better players are more effective at converting White's theoretical advantage
3. **Time Control Preferences:**
    - Stronger players prefer faster time controls (bullet/blitz)
    - This may reflect confidence in pattern recognition and tactical calculation
4. **Data Quality:**
    - The lichess dataset is remarkably clean with no missing values in critical columns
    - The large sample size (500,000 games) provides statistical power for reliable conclusions

8.2 Limitations

- **Sample period:** Only July 2016 data was analyzed
- **Online vs OTB:** Results may differ from over-the-board tournament play
- **Correlation ≠ Causation:** Time control preference may correlate with other factors
- **Rating inflation:** lichess ratings may differ from FIDE ratings

8.3 Next Steps

1. **Machine Learning Models:**
    - Build predictive models for game outcomes using engineered features
    - Test classification algorithms (Logistic Regression, Random Forest, XGBoost)
2. **Opening Analysis:**
    - Analyze win rates by specific openings (ECO codes)
    - Identify openings where White's advantage is strongest/weakest
3. **Time Series Analysis:**
    - Extend analysis to multiple months to identify trends
    - Study rating progression patterns
4. **Network Analysis:**
    - Analyze player matchup networks

    - Identify player communities and performance clusters

5. **Deeper Statistical Tests:**
    - ANOVA for comparing multiple Elo groups
    - Chi-square tests for categorical relationships
    - Effect size calculations (Cohen's d)

Appendix: Technical Information

A1. Environment

- **Python:** 3.x
- **Key Libraries:**
    - pandas (data manipulation)
    - numpy (numerical operations)
    - matplotlib & seaborn (visualization)
    - scipy.stats (statistical testing)
    - statsmodels (proportion tests)

A2. Files Generated

File	Description
`final-project.ipynb`	Main analysis notebook
`README.md`	Project overview for GitHub
`REPORT.md`	This detailed report
`images/*.png`	Visualization exports

A3. Image Catalog

#	Filename	Description
1	`01_white_elo_distribution.png`	Histogram of White player ratings
2	`02_black_elo_distribution.png`	Histogram of Black player ratings
3	`03_base_time_distribution.png`	Distribution of base time values
4	`04_boxplot_all_features.png`	Boxplot of all numeric features
5	`05_boxplot_by_event.png`	Boxplot segmented by event type
6	`06_log_transform_base_time.png`	Base time before/after log transform
7	`07_log_transform_increment.png`	Increment before/after log transform
8	`08_pairplot.png`	Pairplot of key features
9	`09_correlation_with_result.png`	Correlation bar chart with Result

#	Filename	Description
10	`10_h2_basetime_elo_correlation.png`	Scatter plot for Hypothesis 2

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